Muhammet Celal Gokalp

Applications & Systems Engineer | Thermal & Hands-on Engineering

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PROFESSIONAL SUMMARY

Mechanical engineer with an MSc in Sustainable Automotive Engineering and nearly 8 years in automotive systems and applications engineering. My MSc dissertation was on the thermal behaviour of electric machines, including lab testing to build and check a thermal model. I spent 2 years based at a customer site (Jaguar Land Rover) as their main technical contact, and I am used to testing, finding faults and explaining technical issues to customers and suppliers. I understand systems engineering, from requirements to testing. I have not worked on data centre cooling yet, but I learn new areas quickly and I am happy to travel across Europe.

WORK EXPERIENCE

Lead Application Engineer (Solutions & Integration)

10/2021 – Present

FiveAI / Bosch — Cambridge, United Kingdom

- Technical contact between our engineering teams and enterprise customers using our simulation tools for automated driving.
- On the ADA project, a partnership with Volkswagen Group where FiveAI supplied the simulation tooling, trained partner engineers to use our toolchain. This grew into the Simulation EcoSystem solutions team, where I was a founding member. Find and fix tool and integration problems so customers can keep working.
- Plan and run large test campaigns, check results against performance limits, and report issues to the engineering teams.
- Build small internal tools and data pipelines in Python to make testing and reporting faster for the team.

ADAS Systems Engineer (Feature Owner)

11/2018 – 10/2021

AVL Powertrain UK — Coventry, United Kingdom

- Based at the Jaguar Land Rover site for 2 years as AVL's technical contact for driver assistance features (Rear Cross Traffic Alert, Blind Spot Monitor, Rear Pre-Crash).
- Wrote requirements, system state diagrams and test cases from customer specifications (DOORS, MKS Integrity).
- Tested systems in vehicles on the proving ground and on Hardware-in-the-Loop (HiL) and Vehicle-in-the-Loop (ViL) rigs.
- Found faults by reading vehicle network data with Vector CANoe and CANape, then worked with suppliers and engineers to fix them.
- At the LEVC (London Taxi) factory, inspected returned faulty e-machines: multimeter checks, downloading data from proprietary loggers and sending it to developers and suppliers for root cause analysis.
- Presented test results and issues to customer engineering teams and helped them decide on fixes.

Mechanical Engineer Intern

05/2014 – 08/2014

Temsa Global (bus manufacturer) — Adana, Türkiye

- Calculated power and torque needs for a new electric bus powertrain in MATLAB/Simulink, and supported quality work on the production line.

Part-Time Teacher and Program Planner

12/2013 – 01/2017

Space Camp Turkey (associated with NASA) — Izmir, Türkiye

- Taught space technology, physics, chemistry and biology to students aged 9 to 16, explaining technical topics in a simple way.

EDUCATION

MSc in Sustainable Automotive Engineering
10/2017 – 09/2018

Warwick Manufacturing Group, Coventry, UK

- **Grade:** 2:1 (Merit)
- **Dissertation:** Thermal behaviour of electric machines. Ran lab tests to find the variables of a thermal model of a wound stator, and modelled the machine's thermal, mechanical and electromagnetic behaviour (MotorCAD, Portunus, MATLAB/Simulink).

BSc in Mechanical Engineering
08/2011 – 02/2017

Ege University, Izmir, Türkiye

- Covered heat transfer, fluid mechanics and thermodynamics.
- Dissertation: designed, built and controlled a 6-DOF motion platform (SolidWorks, MATLAB, Arduino).

HANDS-ON PROJECTS

Desktop Computers
Ongoing

Personal

- Build my own desktop computers: choose parts, assemble, set up cooling, and fix hardware problems.

Car Repairs
Ongoing

Personal

- Diagnose and repair my own cars at home using workshop manuals and wiring diagrams.

SKILLS

Engineering	Thermal modelling, lab testing, test planning, fault-finding, systems engineering (requirements to testing)
Customer	On-site customer support (2 years at JLR), technical presentations, working with suppliers
Tools	MATLAB / Simulink, MotorCAD, SolidWorks, Python, Vector CANoe / CANape, DOORS, Jira
Languages	English (C2), Turkish (Native)
Other	Full UK driving licence; happy to travel across Europe